



UTM

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APPLICATION DEVELOPMENT

SECTION 01

LAB ASSIGNMENT 2

Industry Project Module Proposal - TVET Attendance Management System

Prototype link : <https://attenda-assist.lovable.app>

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1.0 Project Title

TVET Attendance Management System

2.0 Project Background

The existing attendance management procedure in MARA institutions relies on manual processes and disjointed systems, such as paper-based recording and a series of unconnected digital tools. These methods have proven ineffective, difficult to control, and unable to manage a large number of students.

To overcome these challenges, this project proposes developing a TVET Attendance Management System. The system will transform the manual attendance system into a centralized one that will enhance accuracy, accessibility, and efficiency. It also improves monitoring through reporting and facilitates academic processes such as scheduling and class management.

3.0 Target Users

The system is designed for the following users:

Lecturers — to record attendance and manage class sessions

Admin Staff — to manage timetable and system data

Head of Program — to monitor reports and oversee attendance performance

Students — indirectly involved through attendance tracking

4.0 Mandatory Modules

Login Module:

The login module enables users to securely access the system by verifying their credentials before granting entry. It ensures that only authorized users can use the system and access features according to their roles, which is essential for maintaining data security and system integrity.

Registration:

The registration module allows new users to create an account by providing the required information with proper validation. This ensures that all user data is accurate and complete, supporting reliable user management and smooth system access.

Manage User:

The manage user module enables administrators to view, edit, and remove user accounts within the system, ensuring that user information remains accurate and up to date. It includes role-based access control, allowing different levels of permissions depending on the user's role, so that only authorized individuals can perform sensitive actions.

5.0 Industry Modules

Module 1: Taking Attendance

This module enables lecturers to record student attendance for each class session. Before recording attendance, lecturers select the relevant class, subject, and date. The system offers various attendance statuses: Present, Absent, Medical Leave (MC), and Cuti Kebenaran (CK). Both MC and CK are valid and are counted as present. The module ensures that attendance data is recorded correctly and consistently. This provides a reliable foundation for further analysis and reporting.

Module 2: Report Discipline Issue

This module is designed to help lecturers manage student disciplinary problems efficiently. It allows users to report issues such as frequent absences by students, late attendance, disrespectful behavior, or violations of IKM rules. Each report can include important details such as the student's name, class, date of incident, type of offense, and remarks or evidence. Therefore, this module helps administrators monitor student behavior patterns, identify repeated offenses, and take appropriate action early.

Module 3: Reporting Module

This module allows lecturers and Head of Program to view and analyze attendance records through organized reports by generating summaries such as attendance percentage, total attendance, and absence records based on class, subject, or individual students. The module also provides insights into attendance trends over time, helping users understand student participation patterns presented in a clear and user-friendly format, including simple charts and summaries. This helps lecturers and Head of Program easily identify students with low attendance or irregular patterns.

Module 4: Upload Time Schedule

This submodule allows administrators or authorized staff to upload timetable data into the system. The uploaded data includes class details such as subject, lecturer, time and location. By centralizing timetable information, the system ensures consistency and enables other submodules to function accurately. This eliminates dependency on external scheduling tools and reduces data duplication.

Module 5: Showing Timetable Slot

This submodule displays the lecturer's schedule in an organized and user-friendly interface. Lecturers can easily view their daily or weekly timetable and directly access the attendance taking function from each class slot. This integration improves usability and saves time, as lecturers do not need to navigate multiple systems. It ensures that attendance recording is seamlessly embedded into their daily teaching workflow.

Module 6: Booking Module

This submodule allows lecturers to schedule replacement classes or reschedule existing sessions when disruptions occur so that the teaching and learning sessions are uninterrupted. It allows lecturers to select suitable dates, times and venues while avoiding scheduling conflicts with other classes or resources.

Module	Type	Assigned To
Login	Mandatory	Poh Lok Yee
Registration	Mandatory	Poh Lok Yee
Manage User	Mandatory	Iman Abadi
Taking Attendance	Industry	Tan Zhi Ming
Report Discipline Issue	Industry	Yasmin
Reporting Module	Industry	Poh Lok Yee
Upload Time Schedule	Industry	Sabrina Heng Wei Qi
Showing Timetable Slot	Industry	Sabrina Heng Wei Qi
Booking Module	Industry	Iman Abadi

6.0 Justification

The module distribution was designed based on technical complexity and team members' skill levels. High-complexity modules such as the Attendance System and ST Exam System were assigned to members with strong full-stack experience to ensure system stability and scalability. Medium-level modules such as Promotion and Lecturer Evaluation were assigned to members with foundational knowledge to further develop their skills while contributing meaningfully. Lighter modules were grouped and assigned to a beginner team member to provide a manageable learning curve while ensuring fair workload distribution. Mandatory modules were distributed among experienced members to ensure a solid system foundation. This approach ensures balanced workload, skill growth, and successful project deliver.

7.0 Conclusion

This project aims to develop a comprehensive Attendance Management System that improves the efficiency and reliability of attendance tracking in MARA institutions. By focusing on a single module with detailed submodules, the system ensures depth, quality, and a complete workflow. The proposed solution is expected to enhance academic operations and provide a scalable digital alternative to current manual processes.